

Final RIT Simulation (ALGO2e) — Group Write-up

Group 06

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Case: RIT Algorithmic Market Making: Extensions (CNR, RY, AC)

Account: Group06_main | **Final P&L:** \$2,064,738.33 | **Fines:** \$0.00

(a) What was our strategy going in?

The case is built for market making: post a bid and an ask on CNR, RY and AC, earn the spread plus the limit-order rebate, and keep inventory small. We had already run a passive market maker, and it lost money. It always quoted both sides and leaned *against* price moves. When a stock started trending, our quotes on the losing side kept getting filled, inventory built up in the wrong direction, and the trend ran us over. That is adverse selection: the orders that filled us were the informed ones.

For the final simulation we turned that logic around and ran a **momentum liquidity taker** (`last_simulation.py`):

- **No limit orders.** The bot only sends MARKET orders, so it never has a quote resting in the book to be picked off.
- **Trade with the trend.** For each stock it keeps a 3-tick and a 12-tick exponential moving average of the midquote. The trend signal is their difference, measured in ticks. Fast above slow means go long; fast below slow means go short.
- **Hysteresis.** A position is opened or increased only when $|\text{trend}| \geq 1$ tick, and closed when $|\text{trend}| \leq 0.3$ tick or the trend flips sign. In between, the bot holds. The gap between the two thresholds keeps it from flipping back and forth around zero and paying fees each time.
- **Fee awareness.** Taking costs \$0.0027/share on CNR and \$0.0015 on AC, but on RY you are *paid* \$0.0014/share to take. Adjustments under 500 shares are skipped so small rebalances don't eat the edge.

(b) What changes did we make to handle multiple securities and fees/rebates?

What we saw in practice. In the practice simulations, prices were repeatedly pushed hard in one direction: large, persistent run-ups that looked like market manipulation, on all three stocks and not always at the same time. Our first multi-stock engine was a market maker, and with three stocks it had three times as many ways to be run over: on each stock, the quote that filled was the one on the losing side of the push. So instead of one strategy for the whole book, we ran the same momentum logic *separately on each asset*.

Handling three securities.

- **One signal per stock.** Every second, the script updates a 3-tick and a 12-tick EMA of each stock's own midquote, and measures the trend in ticks of that stock. A push on CNR says

nothing about AC, so each stock enters, holds and exits on its own.

- **Separate state per stock.** Target position, stop-loss (4 ticks per share, then 5s out of that stock) and order pacing (0.3s between orders on the same stock, because RIT reports positions with a lag) are all tracked per ticker.
- **One shared position budget.** The 25,000-share limit applies to the gross position across all three stocks, so each stock is capped at 7,000 shares: $3 \times 7,000 = 21,000 < 25,000$ even if all three trend at once, which is what happened in the final. The script refuses to start if the per-stock cap times the number of stocks exceeds 24,000.
- **Portfolio-level controls.** The drawdown breaker watches the total P&L of the three stocks (\$1,500 below its peak $\Rightarrow$ flatten everything, pause 30s), and all three are flattened in the last 5 ticks.

Handling fees and rebates. Each stock has its own fee schedule (\$ per share):

Ticker	Market order (take)	Limit order (provide)	Structure
CNR	pay 0.0027	earn 0.0023	maker-taker
RY	earn 0.0014	pay 0.0020	inverted
AC	pay 0.0015	earn 0.0011	maker-taker

A market maker would have to treat each stock differently (on RY, every passive fill *costs* money). A momentum taker only pays the taker fee, at most 0.27¢ per share on CNR, and is paid 0.14¢ per share on RY. We gave up the rebates entirely: at 0.11–0.23¢ per share, they are small next to the several-cent inventory losses a market maker took on each push. Because the fees are small next to the 1-tick (1¢) entry gate, we kept one set of rules for all three stocks and controlled fees through trading frequency instead:

- **Hysteresis:** enter at $|\text{trend}| \geq 1$ tick, exit at $|\text{trend}| \leq 0.3$ tick, hold in between, so the bot doesn't flip around zero and pay the fee on every flip;
- **Minimum adjustment:** changes under 500 shares are skipped, so small rebalances don't pay fees for nothing.

(c) What happened during the simulation? Size strategy and script

Ticks 0–210: slow bleed. For most of the session prices were mostly noise around a level, with few lasting trends. A 3/12-tick EMA crossover lags by construction: it buys after a move up, then gets stopped out or exits when the move reverses. Every round trip also paid the spread and, on CNR and AC, the taker fee. P&L drifted down to about $-\$30,000$. That is what a momentum taker should expect when prices tend to revert.

Ticks $\approx$ 210–240: manipulation. Other participants began pushing prices hard in one direction, most likely with very large orders that cleared out the book. To our EMAs this looked like a trend dozens of ticks strong, so the target size went straight to the 7,000-share cap and the bot traded *with* the move as it kept going. P&L jumped from about $-\$30k$ to about $\$2,000,000$ within about 30 ticks (Figure 1). A $\$2M$ gain on at most 21,000 shares ($3 \text{ stocks} \times 7,000$) means prices moved by at least $\$2,000,000/21,000 \approx \95 per share. Normal moves in this case are a few cents, so this jump came from the manipulation, not from our signal having any edge.

Shutting the algorithm off. When the manipulation stopped, we stopped the bot by hand. A pushed price usually snaps back once the pushing ends, and a 3/12 EMA reacts too slowly to get out ahead of that. Running it longer meant either giving back a large part of the gain or, worse, going short into the snap-back. When we stopped, one 7,000-share position was still open (left screenshot, P&L \$2,033,518). It was flat by the end of the case, at a final P&L of \$2,064,738 with zero position and no fines (right screenshot).

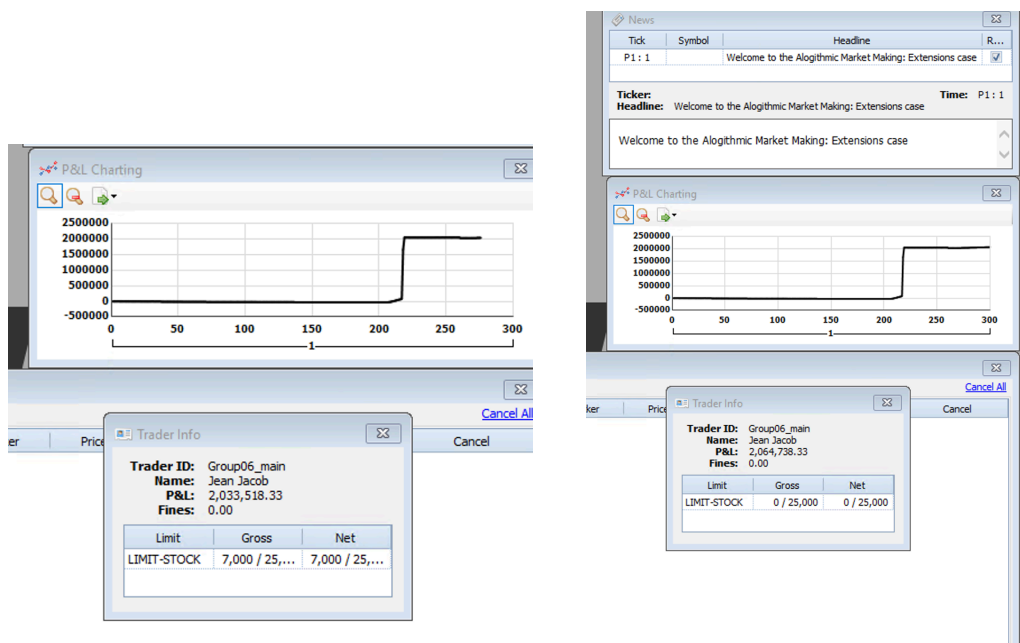


Figure 1: RIT P&L chart and trader info. Left: shortly after the algorithm was stopped, with 7,000 shares still open. Right: end of case, flat.

Size strategy. Position size scales with how strong the trend is:

$$\text{target}_k = \text{sign}(\text{trend}_k) \cdot \min(7,000, 2,000 \times |\text{trend}_k|), \quad \text{rounded down to 100 shares.}$$

A 1-tick trend gives a 2,000-share position; the cap is reached at 3.5 ticks. Holding 7,000 shares in each of three stocks means at most 21,000 shares gross, under the 25,000 limit. That is why we were never fined, even with the manipulated signal maxing out every stock. Orders are capped at 5,000 shares, with at least 0.3s between orders on the same stock, because RIT reports positions with a lag. Risk controls:

- **Per-stock stop:** if the unrealized loss exceeds 4 ticks per share, close the position at market and stay out of that stock for 5 s.
- **Drawdown breaker:** if total P&L falls \$1,500 below its peak, flatten everything and pause for 30 s.
- **End of case:** flatten during the last 5 ticks.

The drawdown breaker turned out to matter most. Once P&L reached \$2M, any \$1,500 pullback would have flattened the book, so the gain was effectively protected.

How to use the script. Enable the API in the RIT client, then run

```
python last_simulation.py --key <API_KEY>
```

(the default host is `http://localhost:7238`). It waits for the case to become ACTIVE, cancels any leftover limit orders, and then trades CNR, RY and AC. It prints a status line every tick and also writes it to `momentum_log.txt`. Each constant can be overridden on the command line, e.g. `--max-position 5000 --ema-fast 2 --ema-slow 8 --tickers CNR,AC`. The script refuses to start if $\text{max-position} \times \text{\#tickers} > 24,000$. Pressing Ctrl+C stops the loop, but it does *not* flatten. Check the RIT client afterwards and close any position left open, which is what happened to us.

MARKET orders versus LIMIT orders

	MARKET (our final strategy)	LIMIT (our market maker)
Guarantee	Execution, not price	Price, not execution
Cost	Pay half the spread plus the taker fee (except RY, which pays a rebate)	Earn the spread plus the rebate
Main risk	<i>Slippage</i> : in a thin or manipulated book, an order can walk several price levels and fill far from the last quote	<i>Adverse selection and inventory</i> : you fill exactly when the price is moving against you, and trends leave you holding the losing side
Other risks	Paying spread and fees on every whip-saw in a mean-reverting market	Non-execution; being picked off on stale quotes; cancel/replace traffic that uses up the API rate limit

In this session the two order types lost money for opposite reasons. The market maker lost because its limit orders gave other traders a free option against it. The taker lost slowly in the quiet part of the session because it paid the spread and fees to follow trends that didn't last. The only reason the taker ended profitable is that the manipulated move was large and one-directional, which is the one environment momentum strategies are built for. MARKET orders were also the right tool for that phase: we needed to get into the move, and later out of it, right away, whatever the fill price. The risk is symmetric, though. The same thin book that let us ride the push up would have filled our exits at extreme prices if we had been on the wrong side. The 5,000-share order cap and the 0.3s gap between orders limited how far a single order could walk the book.

(d) What was our P&L?

Phase	P&L
Ticks 0–210 (normal market)	$\approx -\$30,000$
After the manipulation, algorithm stopped (7,000 shares open)	\$2,033,518.33
End of case (flat, no fines)	\$2,064,738.33

Takeaways. We don't treat the \$2M as evidence that the strategy works. Under normal conditions the momentum taker lost about \$30k. Nearly all of the gain came from other participants manipulating prices, which would be illegal in a real market and in any case can't be repeated. What we did get right was the risk management around that event: hard position caps kept us inside the limits (no fines), and turning the algorithm off as soon as the manipulation ended locked in a gain that a lagging trend signal would otherwise have given back. If we were to keep a strategy

for normal conditions, it would be the market maker with a trend filter: quote passively to earn the spread and rebate, but pull or skew quotes when the EMA trend signal shows a strong move.